Anatomy of Associative Recall in Fixed-State Recurrences: A Matched-State Decomposition, an Interference Wall, and a Curriculum That Breaks It

Julian Boesch[1] Andrew Wee[1]

Abstract

Fixed-state recurrences—linear attention and state-space models—are reported to lag attention on associative recall, but whole-architecture comparisons cannot say which ingredient is responsible. We decompose masked multi-query recall at a fixed state budget along three single-knob axes: a short causal convolution, the state-transition structure (rank-1 delta rule vs. diagonal), and decay. The convolution dominates ($\approx$+0.5 recall in both families under matched training): comparisons that pit convolution-free cells against a convolution-equipped Mamba measure the missing convolution, not the recurrence. The rank-1 transition adds a within-cell margin over its diagonal ablation (+0.19/+0.32 at K=16/32) that shrinks to +0.03 once both cells carry the convolution, and a state-matched Mamba-2 still ties or beats the unarmed rank-1 cell: no class claim survives. Load and distance then dissociate. Cells that solve 32-pair recall degrade gracefully with load but fail at chance retrieving 4 pairs across a distractor haystack—at the shortest tested length, flat across lengths, identically across transitions. This is interference under sparse supervision, not capacity: a table-to-query distance curriculum takes the unchanged architecture from 0.021 to 1.000. Training in this regime is a lock-in lottery: the curriculum raises the lock-in rate rather than the mean, and a shaped ramp is the one schedule contrast that clears significance (0/9 uniform vs. 4/5 at L=256). Bidirectional denoiser cells read the query before the haystack natively, but at ten seeds this confers no measurable advantage over causal training; any circuit solving collision-key retrieval needs two layers. Arming the cell for recall costs nothing on an S_5 state-tracking guardrail. These replace "recurrent models are bad at recall" with a measured decomposition, a discriminating diagnosis, and two cheap interventions.

[1]Obit Development. Correspondence to: Julian Boesch <jboesch@obitmc.com>, Andrew Wee <awee@obitmc.com>.

Preprint. Preliminary results, shared for discussion; not peer reviewed.

1. Introduction

Associative recall—binding key–value pairs in context and retrieving a value when its key reappears—is the capability axis on which fixed-state sequence mixers most visibly trail attention. A line of work has made this precise: synthetic multi-query associative recall (MQAR) separates attention from efficient mixers and predicts much of the language-modeling gap between them (Arora et al., 2023; 2024a), and recall remains the standard lens on what a bounded recurrent state can and cannot retain (Jelassi et al., 2024; Hsieh et al., 2024).

But a modern recurrent cell is not one mechanism. A Mamba-2 block, for instance, is a diagonal selective state transition plus an input-dependent gate plus a short causal depthwise convolution (Gu & Dao, 2023; Dao & Gu, 2024); a Gated DeltaNet block is a rank-1 delta-rule transition plus a learned decay (Yang et al., 2024b;a); RWKV-7 couples a vector decay with a rank-1 removal term (Peng et al., 2025). Whole-architecture comparisons therefore confound at least three design axes, and the field's summary judgments ("DeltaNet-style cells recall well," "Mamba recalls better than RNNs," "gating helps") average over knobs that can be toggled independently. Recent causal-intervention evidence sharpens the concern: Mamba's induction behavior appears to live in its short convolution rather than its state-space scan (Arora et al., 2025; Parnichkun et al., 2025), though this attribution is itself contested once learning rates are tuned (Okpekpe & Orvieto, 2025). What is missing is a controlled decomposition: all knobs, one harness, one fixed state budget.

This paper contributes that decomposition, and then uses it to re-diagnose two failure modes that the decomposition alone does not explain.

Contributions.

1. A matched-state decomposition of recall, and the confound it resolves (§4). At a fixed recurrent-state budget (1,024 state elements; ∼29k parameters) we toggle, one knob at a time: short convolution × transition structure (rank-1 vs. diagonal) × decay. The convolution is the dominant lever and transfers across cell families (+0.55 delta-rule, +0.47 diagonal at K=32), corroborating intervention evidence that Mamba's recall is convolution-borne (Arora et al., 2025; Parnichkun et al., 2025); we also measure the learning-rate sensitivity that Okpekpe & Orvieto (2025) raise. The rank-1 transition contributes an independent, ablation-grade margin over its own diagonal ablation (+0.19/+0.32 at K=16/32, 10 seeds, rebinding-controlled), which shrinks to +0.034 (p=0.0023) once both cells carry the convolution; decay's apparent −0.32 recall cost does not survive 10-seed hardening (n.s.). The correction this forces: comparisons of convolution-free recurrent cells against convolution-equipped Mamba mismeasure the transition; armed with the same convolution, a gated delta-rule cell reaches 0.99 at the hardest matched-state setting. The converse also holds: a state- and parameter-matched Mamba-2 ties or beats the unarmed rank-1 cell, so no class claim ("rank-1 > diagonal") survives.

2. Capacity vs. distance, discriminated (§5). Load (K) produces graceful degradation; distance across a distractor haystack produces a wall: chance already at the shortest tested length—inside the training distribution—identical across delta-rule, selective-diagonal, and RWKV-7 transitions, while a 2-layer attention control solves the task at 1.0. This signature, completed by the curriculum result below, separates interference under sparse supervision from both capacity accounts and decay accounts (Sridhar & Johansen, 2026) of recurrent retrieval failure.

3. The wall is a training-coverage gap, and training in this regime is a lock-in lottery (§6). With the architecture unchanged, a table-to-query distance curriculum lifts the walled cell from 0.021 (chance) to 1.000—an existence proof by construction. Five-seed replication reframes the attribution as a rate shift: lock-in 1/5 (dense-only) → 3/5 (curriculum) → 4/5 (both), directionally consistent, pairwise n.s. at n=5—and dense-only training solves the task outright on one seed. The fix transfers unchanged to 16× state (d=128); at L=256 the uniform curriculum fails everywhere (0/9 trials) but a shaped ramp locks in 4/5 seeds (p≈0.005, Fisher)—the one curriculum-shape contrast that clears significance.

4. Bidirectional denoisers read twice for free—architecturally; no advantage is measured (§7). The backward stream of a bidirectional masked-denoiser cell sees the query before the haystack, the prefix-encoder property that Just-Read-Twice engineers into recurrent LMs (Arora et al., 2024b). A collision-key discriminator (decoys reuse the table's keys, defeating any lexical write gate) tests whether the property confers an advantage. At ten seeds it does not: collision training is a bimodal lock-in lottery (bidirectional 3/10 seeds solve, causal 1/10; mean difference n.s., p=0.63), and shaped-curriculum training that stabilizes the lottery (9/10 lock-in) does so at exact directional parity. What is robust: both directions can solve the task; any solving circuit needs two layers (a mark-and-route circuit—the one-layer version collapses); the convolution-free cell never locks in even under the shaped curriculum (0/10 vs. 9/10 on the identical recipe, $p≈10^{-4}$); and curriculum shape is a powerful direction-agnostic stabilizer (1/10 → 9/10, $p≈6×10^{-4}$).

5. A no-cost guardrail (§8). Arming the cell for recall does not measurably hurt S_5 state-tracking learnability at any tested depth.

All experiments run on a deliberately small, kernel-free, commodity-GPU harness (pure-PyTorch reference cells, a state-matched pure-PyTorch Mamba-2 comparator, and an atomic-free two-pass Triton backward that runs on pre-Volta hardware; Appendix A), which is what makes single-knob matched-state factorials cheap to replicate. The price is scale: every result here is toy-scale ($d \in \{32, 128\}$, synthetic tasks), and §9 states the resulting limits explicitly.

2. Related Work

Recall in efficient mixers. MQAR and its relatives were introduced to explain the attention–SSM gap (Arora et al., 2023), and recall-throughput tradeoffs drove a generation of linear-attention designs (Arora et al., 2024a; Katharopoulos et al., 2020; Schlag et al., 2021). The delta rule (Yang et al., 2024b) and its gated variant (Yang et al., 2024a) explicitly target recall via key-conditioned replacement; RWKV-7 generalizes the transition to diagonal-plus-rank-1 (Peng et al., 2025). Our contribution to this line is not a new cell but a controlled decomposition of existing ingredients at matched state, including the negative result that the rank-1 advantage over a diagonal ablation does not extend to a class advantage over Mamba-2.

Where Mamba's recall comes from. Causal interventions locate Mamba's induction in its short convolution (Arora et al., 2025); effective-state-size analyses likewise find many linear structures collapse on recall without their convolutions (Parnichkun et al., 2025). Okpekpe & Orvieto (2025) complicate the attribution, showing tuned learning rates can recover much of Mamba's recall without the convolution—a learnability confound. Our factorial speaks to this debate with a design the intervention studies lack (single-knob toggles at fixed state, both cell families) but at toy scale; we find the convolution's contribution large, additive, and family-transferable under matched training, and we measure the learning-rate sensitivity directly (§4.2).

Long-context retrieval failures. That SSM-family models fail needle-in-a-haystack retrieval is well documented (Hsieh et al., 2024; Jelassi et al., 2024; Ben-Kish et al., 2024); proposed accounts include state capacity and collapse (Chen et al., 2024), finite-horizon state decay (Sridhar & Johansen, 2026), and under-explored state distributions at unseen lengths (Buitrago Ruiz & Gu, 2025). Our haystack wall differs from these accounts in its signature—chance at the shortest tested length and flat across lengths, within the training distribution, transition-independent—and in its resolution: a training-signal change alone removes it (§6), which no purely structural account predicts.

Training-side fixes. Birdie improves SSM retrieval with reward-driven objectives and mixed training procedures (Blouir et al., 2024); Okpekpe & Orvieto (2025) and Buitrago Ruiz & Gu (2025) frame recurrent retrieval and length generalization as learnability problems; curricula are classical (Bengio et al., 2009). We add the sharpest version of the claim for retrieval distance: a chance→perfect existence proof with a single-axis curriculum on an unchanged architecture, its lever attribution (2×2), and its failure boundary in length.

Recurrent denoisers for diffusion LMs. Masked discrete diffusion (Austin et al., 2021; Sahoo et al., 2024) makes the denoiser bidirectional by construction; bidirectional Mamba denoisers match Transformer DLMs (Singh et al., 2025), and RWKV-backbone block diffusion exists (Lin et al., 2026). Just-Read-Twice (Arora et al., 2024b) showed causal recurrent LMs recover most of the recall gap when the query precedes the context. Our contribution is the explicit connection: bidirectional denoisers possess the JRT property natively—no prompt repetition, no bespoke prefix-LM—together with the depth requirement of any circuit exploiting it (two layers: mark-and-route) and its apparent local-binding requirement (the short convolution). Whether the native property confers a measurable advantage over matched causal training is open: our ten-seed collision replication finds no significant directional difference (§7).

State tracking. Diagonal SSMs and attention are limited on group-composition state tracking in ways rank-1-transition recurrences need not be (Merrill et al., 2024; Grazzi et al., 2025; Liu et al., 2023; Delétang et al., 2023). We use an S_5 word problem only as a guardrail (does arming for recall cost state-tracking learnability?) and make no expressivity-class claims here.

3. Cells, Tasks, and Protocol

3.1. The cell zoo

All cells are implemented in one harness as sequence mixers with identical embedding, readout, and training loops; we call each cell configuration under test an arm. Per head, a matrix state $\mathbf{S}_t \in \mathbb{R}^{d_v \times d_k}$ is updated by a cell-specific transition; the output is a state readout. We write k_t, v_t, q_t (or r_t) for the per-token projections.

DeltaNet (rank-1, no decay) applies the delta rule (Schlag et al., 2021; Yang et al., 2024b), with input-dependent write strength $\beta_t \in (0, 1)$:

$$\mathbf{S}_t = \mathbf{S}_{t-1}\big(\mathbf{I} - \beta_t k_t k_t^\top\big) + \beta_t\, v_t k_t^\top, \quad o_t = \mathbf{S}_t q_t. \tag{1}$$

Gated DeltaNet (gated_deltanet; rank-1 + decay) adds a learned scalar gate $\alpha_t \in (0, 1)$ (Yang et al., 2024a):

$$\mathbf{S}_t = \alpha_t\, \mathbf{S}_{t-1}\big(\mathbf{I} - \beta_t k_t k_t^\top\big) + \beta_t\, v_t k_t^\top. \tag{2}$$

RWKV-7 reference (dg_rank; rank-1 + vector decay) uses the diagonal-plus-rank-1 transition (Peng et al., 2025):

$$\mathbf{A}_t = \mathrm{diag}(w_t) - (a_t \odot \hat{\kappa}_t)\hat{\kappa}_t^\top, \tag{3}$$

$$\mathbf{S}_t = \mathbf{S}_{t-1}\mathbf{A}_t + v_t k_t^\top, \qquad o_t = \mathbf{S}_t r_t, \tag{4}$$

with data-dependent vector decay $w_t \in (0, 1)^{d_k}$, removal key $\hat{\kappa}_t$, and in-context rate a_t. Its diagonal ablation (dg_diag) sets the rank-1 term to zero ($a_t \equiv 0$), leaving $\mathbf{A}_t = \mathrm{diag}(w_t)$: the same cell, same parameter count, one knob.

Mamba-2 reference (mamba2_ref) is a pure-PyTorch state-matched implementation of the Mamba-2 SSD recurrence (Dao & Gu, 2024): a per-head scalar selective diagonal scan over an explicit state $\mathbf{H}_t \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{state}}}$,

$$\mathbf{H}_t = a_t\, \mathbf{H}_{t-1} + x_t\, b_t^\top, \qquad y_t = \mathbf{H}_t\, c_t, \tag{5}$$

with input-dependent (a_t, b_t, c_t) and the standard short causal depthwise convolution on its input projections. mamba2_ref_noconv sets the convolution width to 1 (a true toggle). Because this comparator needs neither Triton nor mamba_ssm, the full factorial runs on commodity GPUs; its cost is that it undertrains relative to the official fused kernel (§9).

The arming knob. A width-4, strictly causal (left-padded), depthwise convolution on the interaction projections—(q, k, v) for the delta-rule cells, (v, k, r) for the RWKV-7 cells—applied pre-activation exactly as in Mamba-2. The convolution adds ≈0.5k parameters and zero recurrent state, so state-matching is preserved. armed_gdn ≔ gated_deltanet + this convolution.

Attention control. A standard bidirectional (or causal, where noted) softmax-attention block (Vaswani et al., 2017), used as the ceiling for the haystack task (§5) with widened heads (head dimension 16). At the constrained-width factorial setting (d=32) no valid attention control exists at matched width (the head dimension degenerates, and a two-head variant collapses at one layer for a structural reason; §9), so attention is not reported there.

Bidirectional variants. For §7, a cell is made bidirectional in the standard masked-denoiser way (Singh et al., 2025): a paired backward stream runs the same recurrence on the flipped sequence and the two streams are merged per position. armed_gdn_fwd denotes the causal (forward-only) ablation.

3.2. State matching

The factorial is run at constrained state, where recall must be discriminative: d=32, one layer. The rank-1 cells carry $d_k^2 = 32^2 = 1024$ recurrent state elements per head group; the Mamba-2 reference is set to d_{state}=16 so that $d_{\text{inner}} \cdot d_{\text{state}} = 64 \cdot 16 = 1024$ elements—matched state, and matched parameters (∼28.9k vs. ∼29.3k). At ample state (d=128) every cell solves every setting and nothing discriminates, which is itself worth reporting: recall differences among modern cells are a constrained-state phenomenon.

3.3. Tasks and floors

Masked MQAR. A table of K key–value pairs followed by queried keys whose answers are single masked tokens; the model is trained to fill the masks. Values are drawn fresh at random per sequence from a 26-token range, so no key→value prior exists to memorize. $K \in \{8, 16, 32\}$. Floors: the value-marginal is $1/26 \approx 0.038$; the strongest no-binding strategy (emit a random table value) scores ≈0.068. In the hardened protocol we also report the table floor $1/K$ (guess among the K values actually present), the floor the naive marginal comparison misses.

Haystack retrieval. 4 key–value pairs, then a distractor haystack of task-format tokens, then the query; the answer is a single masked token at the final position; sequence lengths $L \in \{64, \ldots, 512\}$. Chance is $1/54 \approx 0.019$. The query-at-end layout makes the task 2-hop; all arms get 2 layers and a proper training budget (the 1-layer version fails for every arm including attention, an under-capacity artifact we exclude).

Collision-key variant. As above, but decoy pairs in the gap reuse the table's own keys with fresh random values; ground truth is the first (table) binding. This defeats any lexical write gate: distractors are lexically indistinguishable from table keys. Floors: value-marginal $1/54 \approx 0.019$; the strongest non-binding positional strategy (emit a random table-position token) scores ≈0.26.

S_5 guardrail. A running-product word problem over the symmetric group S_5: generators are drawn uniformly from the full group (so the answer marginal equals $1/|G| = 1/120 \approx 0.008$ at every depth), each group element is a single token, and all product slots are masked simultaneously (one parallel fill), so no answer token is completable from a visible prefix. Composition depths $L \in \{2, 4, 8, 16\}$ at d=128. These three design choices matter: they make the probe structurally immune to the answer-prior and prefix-leakage confounds that can silently invalidate state-tracking probes with multi-token, teacher-forced answers.[1]

3.4. Statistical protocol

Exploratory sweeps use 3 seeds. Headline ablations are hardened: 10 seeds; two-sided permutation tests on mean gaps; and a rebinding control—after train-

[1] A companion report documents that failure anatomy in a conversion-evaluation setting; this paper's probes are immune by construction, and we state floors with every result.

ing, evaluate the same model on sequences whose key–value bindings have been deranged, scoring the original value. A genuine retriever's rebinding score must collapse toward the floors (it emits the rebound value, not the trained one); a value-salience heuristic does not. These rules operationalize, for recall and state-tracking probes, general lessons on prefix leakage in multi-token targets, answer-prior baselines, and control tasks (Bachmann & Nagarajan, 2024; Holtzman et al., 2021; Hewitt & Liang, 2019). Where seed outcomes are bimodal (a seed either "locks in" or does not), we report lock-in rates (fraction of seeds above threshold) with Fisher exact tests rather than means, since mean±std is meaningless for bimodal outcomes. Every table caption states its seed count, and every task quotes its floors (derivations in Appendix C).

4. A Matched-State Decomposition of Recall

4.1. The factorial

Table 1 shows the constrained-state fight at the hardest load, K=32.[2] Table 2 gives the full K-sweep (72/72 trials).

4.2. Reading the factorial: three separable levers

Figure 1 shows the three toggles side by side.

Axis 1: the convolution ($\approx +0.5$, family-transferable). Holding the recurrence fixed and toggling only the convolution: the delta-rule cell moves 0.446 $\rightarrow$ 0.993 (+0.55; gated_deltanet $\rightarrow$ armed_gdn) and the diagonal cell moves 0.172 $\rightarrow$ 0.641 (+0.47; mamba2_ref_noconv $\rightarrow$ mamba2_ref). Both legs are true within-cell toggles. The convolution is a state-free local primitive—it adds no recurrent state—yet it is the largest single lever in the factorial, of similar magnitude in both families. This is consistent with, and adds a controlled factorial to, intervention evidence that Mamba's induction is convolution-borne (Arora et al., 2025; Parnichkun et al., 2025). We also ran the learning-rate check of Okpekpe & Orvieto (2025) in our own harness (no-conv arms at $\{3\text{e-}4, 1\text{e-}3, 3\text{e-}3\}$, 5 seeds each, rebinding-controlled): per-arm tuning rescues the delta family but not our diagonal cell. Tuned no-conv gated_deltanet reaches 0.870 at lr 3e-3 (vs. 0.501 matched), while mamba2_ref_noconv peaks at 0.172 across the sweep, leaving its within-cell conv gap (+0.42) intact under tuning. Re-tuning the armed arms as well: at lr 3e-3 all three saturate (armed_gdn 0.999, armed rank-1 1.000, armed diagonal 0.997; 5 seeds each), so the tuned-vs-tuned delta-family conv gap is $\approx$+0.13 (0.999 vs. 0.870), and the with-conv transition margin of +0.034 is a matched-lr statement (at tuned lr both armed cells sit at ceiling, margin $\approx$0.002). The conv's contribution is therefore optimization-conditional in the delta family (large matched, $\approx$+0.13 tuned) and robust in the diagonal cell; magnitudes should always be quoted with their tuning regime.

Axis 2: the transition ($\approx +0.3$ without the convolution; $\approx +0.03$ with it). The clean evidence is the within-cell pair: dg_rank vs. dg_diag—one architecture, one parameter count, one switch (the rank-1 term on or off). At 3 seeds, no conv: 0.639 vs. 0.359 (+0.28). The hardened 10-seed rerun (§4.3) confirms it with controls: +0.191 at K=16 (p=0.0002), +0.319 at K=32 ($p<10^{-4}$). The with-conv quadrant is ablation-grade as well (same protocol and harness as Table 1, 10 seeds, rebinding-controlled): the convolution-equipped pair gives 0.992 (rank-1) vs. 0.958 (diagonal), +0.034 (p=0.0023) at K=32; at K=16 all armed arms sit at ceiling ($\geq$ 0.9998, 10 seeds), so the margin is a highest-load phenomenon. The transition's contribution is therefore conditional on the convolution: large when the cell lacks local mixing, an order of magnitude smaller—though still resolvable—once the convolution is present and both cells operate near ceiling.

Axis 3: decay (-0.32 at 3 seeds; does not survive hardening). Apples-to-apples within the delta-rule family at 3 seeds: deltanet (no decay) 0.769 vs. gated_deltanet (decay) 0.446 at K=32. At 10 seeds (same protocol as Table 1) the effect does not hold: 0.637 vs. 0.501, +0.135, p=0.14 (n.s.), with the no-decay cell severely seed-bimodal (per-seed range 0.098–0.927). Decay's recall cost at constrained state is suggestive but unresolved at this power; what is settled is that armed_gdn wins with its decay in place, on the strength of the convolution plus the delta rule. The converse claim—that decay buys state-tracking—is likewise not established (a 10-seed lock-in comparison of gated_deltanet vs. deltanet on S_5 is directionally positive but non-significant, p=0.65). We also note boundary conditions from the literature: fine-grained (channel-wise) gating can improve recall over scalar gating (Kimi Team, 2025), so any liability measured here is specific to the scalar-decay knob at constrained state.

[2] No attention control is reported at this setting: at d=32 with 8 heads the head dimension degenerates to 4, and a two-head variant collapses at one layer because MQAR is an induction-head task that needs two attention layers (§9). Attention appears as a valid ceiling in the two-layer haystack experiments (§5), with head dimension 16.

Table 1. Constrained state, K=32 (hardest setting). Masked recall, mean over n=3 seeds with per-seed values. State-matched throughout (1,024 elements). Floors: value-marginal ≈ 0.038; no-binding ≈ 0.068. For context: a 3-seed baseline with the official fused Mamba-2 kernel on the same task protocol, run in a separate session, measured 0.884 at the state-matched d_{state}=16 and 1.000 at d_{state}=64 (§9).

arm (K=32, d=32)	mean recall	per-seed	ingredients
armed_gdn	0.993	0.98 / 1.00 / 1.00	rank-1 + decay + conv
deltanet	0.769	0.63 / 0.75 / 0.93	rank-1, no decay, no conv
mamba2_ref (state-matched)	0.641	0.48 / 0.66 / 0.78	diagonal + selectivity + conv
dg_rank (RWKV-7 rank-1)	0.639	0.74 / 0.54 / 0.63	rank-1 (vector decay), no conv
gated_deltanet	0.446	0.44 / 0.57 / 0.32	rank-1 + decay, no conv
dg_diag (RWKV-7, rank off)	0.359	0.37 / 0.31 / 0.39	diagonal (vector decay)
mamba2_ref_noconv	0.172	0.18 / 0.14 / 0.19	diagonal + selectivity, no conv

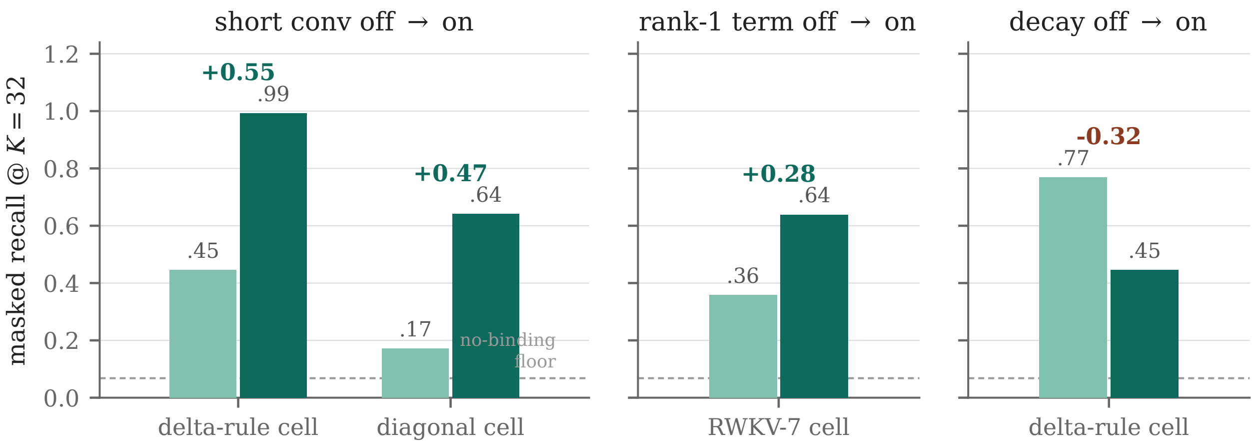

Figure 1. The three single-knob toggles at K=32 (constrained state; n=3 seeds; values from Tables 1 and 2). Light bars: ingredient off; dark bars: on. The short convolution is the dominant, family-transferable lever; the rank-1 transition adds an ablation-grade within-cell margin (hardened at 10 seeds: +0.319 without the conv, +0.034 with it; Table 3, §4); the decay bar's apparent cost does not survive 10-seed hardening (n.s., §4). Dashed line: strongest no-binding floor (≈ 0.068).

Table 2. Full K-sweep, mean masked recall over n=3 seeds (72/72 trials). Everything solves K=8; discrimination grows with load; the no-conv diagonal fails by K=16; the graceful decline of the unarmed arms from K=8 to K=32 is the capacity signature—contrast the haystack wall of §5.

arm	K=8	K=16	K=32
armed_gdn	1.000	1.000	0.993
deltanet	1.000	1.000	0.769
dg_rank (RWKV-7)	1.000	0.998	0.639
mamba2_ref (state-matched)	1.000	0.758	0.641
gated_deltanet	1.000	0.970	0.446
dg_diag	1.000	0.835	0.359
mamba2_ref_noconv	0.582	0.230	0.172

4.3. The hardened transition ablation, with controls

Table 3 reports the 10-seed, rebinding-controlled rerun of the transition axis. Three findings:

1. The within-cell ablation replicates decisively. dg_rank > dg_diag: +0.191 at K=16 (p=0.0002), +0.319 at K=32 ($p<10^{-4}$), gaps far larger than seed variance, and the rebinding control verifies that what is being measured is binding. The advantage is delta-rule-general, not RWKV-7-specific: deltanet ≈ dg_rank at K=32 (p=0.36), both ≫ dg_diag.

2. No class claim survives. A state- and parameter-matched Mamba-2 is a diagonal recurrence, and it ties the rank-1 cell at K=16 (p=0.25) and beats it at K=32 (0.849 vs. 0.702, p=0.0036). "Rank-1 class > diagonal class" is false in this data. The diagonal ablation is a crippled member of its class, not a representative; the correct statement is: within one cell at matched parameters, the rank-1 transition adds recall capacity per state dimension.

3. The two statements together resolve the folklore. The reason convolution-free recurrent cells lose to

Table 3. Hardened transition ablation (n=10 seeds; d=32, N=1; the three delta-rule arms are convolution-free, while the mamba2_ds16 comparator is a state-matched Mamba-2 that retains its stock short convolution). "Rebound" is accuracy on the rebinding control (bindings deranged, original value scored). At K=16 every arm falls to or below the table floor ($1/K$=0.062); at K=32 rebound sits at 0.042–0.073—slightly above the $1/K$=0.031 floor but near the value-marginal (0.038) and an order of magnitude below trained accuracy. Models emit the rebound value: the recall is genuine key→value binding, not value salience. Permutation tests are two-sided on mean gaps; gaps and drops are computed on full-precision values, not the rounded cells shown.

K	arm	mean [95% CI]	lock-in ≥ 0.9	rebound	drop
16	dg_rank (rank-1)	0.998 [0.995, 0.999]	10/10	0.024	+0.974
16	dg_diag (diagonal ablation)	0.806 [0.721, 0.892]	3/10	0.038	+0.769
16	deltanet (rank-1)	1.000 [0.999, 1.000]	10/10	0.024	+0.976
16	mamba2_ds16 (diagonal, matched)	0.999 [0.999, 1.000]	10/10	0.024	+0.975
32	mamba2_ds16 (diagonal, matched)	0.849 [0.807, 0.890]	3/10	0.042	+0.807
32	dg_rank (rank-1)	0.702 [0.631, 0.769]	0/10	0.051	+0.651
32	deltanet (rank-1)	0.623 [0.482, 0.741]	1/10	0.057	+0.566
32	dg_diag (diagonal ablation)	0.382 [0.322, 0.450]	0/10	0.073	+0.309

Mamba on recall is the convolution, not the recurrence (Axis 1); and the reason the rank-1 transition looked alternately strong and weak across studies is that its genuine within-cell effect was being compared across architectures against cells that carry the other two knobs.

The armed arms at 10 seeds (rebinding-controlled, same harness as Table 1). armed_gdn 0.983 (per-seed min 0.917; rebinding 0.037, at the value-marginal floor), armed RWKV-7 (dg_rank + conv) 0.992 $[0.970, 0.999]$, armed diagonal ablation 0.958 $[0.870, 0.996]$. Within-harness, armed_gdn clears mamba2_ref by +0.392 (p=10^{-5}), and the two armed rank-1 families are statistically indistinguishable (p=0.37)—once armed, the delta-rule cell family does not matter. That an armed diagonal cell reaches 0.958 is the factorial's sharpest single number: at constrained state, the convolution is very nearly the whole recall story.

The armed cell vs. real Mamba. armed_gdn at 0.983 (10 seeds) clears the official fused Mamba-2 at matched state (0.884, measured in a separate session with the same probe) by ≈+0.10; within our harness it clears the under-trained mamba2_ref by +0.39. A paired, per-arm-tuned measurement of the official kernel in one session remains to be done (§9).

5. Capacity Curves vs. the Interference Wall

The K-sweep (Table 2) shows what a capacity limit looks like: everything solves K=8; the unarmed arms decline gracefully to K=32. The haystack task shows something categorically different—total, immediate, and length-flat (Table 4); Figure 2 juxtaposes the two axes.

Table 4. The haystack wall, flat in length (n=5 seeds per cell, means shown; 2 layers, proper budget; chance ≈ 0.019). The same recurrences that store 32 competing pairs at ≥ 0.99 (Table 2) fail at chance on 4 pairs across a distractor haystack—already at the shortest tested length, for every transition type, with no trend through 4× the length: all 45 recurrent trials lie in $[0.014, 0.023]$, straddling chance. The properly headed attention control is exact at every length.

arm (2 layers)	L=64	L=128	L=256
attention (head dim 16)	1.000	1.000	1.000
armed_gdn	0.018	0.018	0.019
mamba2_ref	0.017	0.019	0.019
dg_rank (RWKV-7)	0.018	0.018	0.018

Four properties discriminate the failure mode:

1. It is not capacity. The same cells store 32 competing bindings at ≥ 0.99 (Table 2) but fail on 4 bindings plus distractors. The state is 8× undersubscribed relative to demonstrated capacity.
2. It is a wall, not a dip. The failure is already total at the shortest tested length, inside the training distribution, at table-to-query gaps of only tens of tokens—and it stays exactly there through 4× the length: means 0.017–0.019 across $L \in \{64, 128, 256\}$, every recurrent trial within $[0.014, 0.023]$ (Table 4). A capacity limit degrades gracefully (as K does); a decay account (Sridhar & Johansen, 2026) predicts degradation that grows with distance—not chance at minimal distance, and not a floor with no length gradient. There is no distance trend for state decay to explain: the binding is never learned, not progressively lost. (A 1-layer sweep over L=64–512 was likewise flat at chance but is excluded as under-capacity for this 2-hop

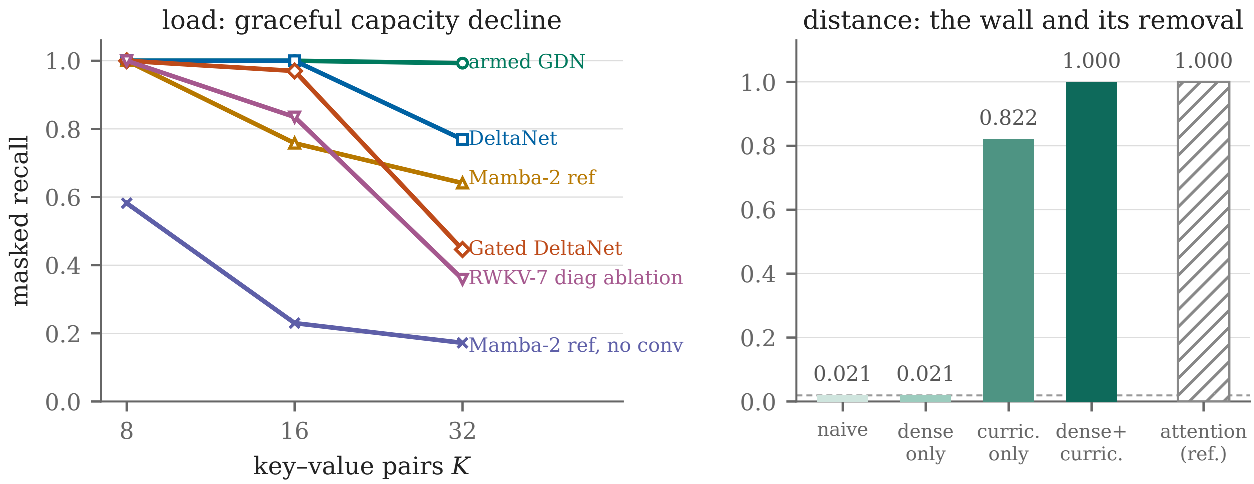

Figure 2. Load vs. distance. Left: the K-sweep (Table 2; n=3 seeds)—a graceful, capacity-style decline; the RWKV-7 rank-1 arm (0.639 at K=32) is omitted for legibility. Right: the haystack wall and its removal for the same bidirectional armed cell at L=64 (Tables 4 and 5; bars show seed-0 accuracy; Table 5 gives the n=5 lock-in rates): naive training sits at chance (dashed line, $\approx$ 0.019); the distance curriculum unlocks the task. Hatched bar: the attention reference on the fixed layout (§6 explains why no attention ceiling is reported under the curriculum itself).

task.)

3. Transition richness does not help. GDN's delta rule, Mamba's selectivity, and RWKV-7's removal key wall identically—at chance at every tested length (n=5 each). Whatever is failing is not the state-transition's expressivity.

4. Attention is untouched, as expected: per-token KV entries cannot be overwritten by later writes—and the control is exact at every tested length (15/15 trials at 1.000).

The diagnosis consistent with all four: interference under sparse supervision. The model must learn a selective-write policy (store table pairs; ignore distractors), but the training signal—one masked token at the end of the sequence—gives the write gate almost no gradient. Distractor writes overwrite the bindings before the query arrives. This account differs from decay-based explanations of SSM retrieval failure (Sridhar & Johansen, 2026): a decay account predicts distance-dependence and does not predict that a pure training-signal change can fix the failure at fixed architecture. Section 6 runs exactly that test.

6. The Wall Is a Training-Coverage Gap

Same task, same architecture, same budget as Table 4; only the training signal changes. Dense supervision: N distinct queried pairs per sequence, all supervised. Distance curriculum: the table→query gap is sampled uniformly in $[0, \max]$ per batch; evaluation is always at max gap. (Exact flags in Appendix B.)

The existence proof. The headline row is an existence proof by construction and needs no seed band to be valid as an existence claim: an unchanged 2-layer fixed-state recurrence that sat at chance (0.021) under naive training reaches 1.000 on the same task under a distance curriculum. The wall of §5 is therefore a training-coverage gap, not an architectural limit—for this task, at this length. This is the sharpest form of a conclusion that the recent literature reaches by other routes: recurrent retrieval failure is substantially a learnability problem (Okpekpe & Orvieto, 2025; Buitrago Ruiz & Gu, 2025; Blouir et al., 2024).

Does it survive off the bench? The curriculum is the one result here that makes a deployment-shaped claim, so it is the one we took to a real converted LM. That work—staged distillation of an attention teacher into a recurrent bidirectional diffusion student, at 1.7B and 8B—is reported in a companion paper (Boesch & Wee, 2026), and we summarize only its bearing on the claims above. It replicates the phenomenon rather than the number: at conversion scale the curriculum's effect is again a lock-in rate. Closing the loop on the curriculum—advancing the gap cap only while retrieval accuracy holds—raises that rate from one partial lock-in in three to 3/3 at 0.94–0.99, and the recipe carries to 8B (2/3 seeds) with recall transfer rising from 0.000 to a mean of 0.901. Two of this section's conclusions therefore hold at four orders of magnitude

Table 5. Curriculum lever attribution, replicated at n=5 seeds (L=64 haystack, 2 layers). Training in this regime converges stochastically (§7), so we report lock-in rates (seeds > 0.8) with per-seed values. The chance→1.000 headline is an existence proof by construction and does not rest on seed count; the attributions are rate shifts, pairwise n.s. at n=5 (e.g., 4/5 vs. 1/5: $p\approx0.21$, Fisher).

cell (L=64, 2 layers)	dense	curric.	lock-in	per-seed
armed_gdn (bidir)	4	✓	4/5	1.00/0.87/0.88/0.99/0.36
armed_gdn (bidir)	1	✓	3/5	0.82/0.13/0.06/1.00/0.92
armed_gdn (bidir)	4	—	1/5	0.02/0.02/0.02/1.00/0.02
armed_gdn_fwd (causal)	4	✓	3/5	0.84/0.75/0.96/1.00/0.06
attention (abs. pos.)	4	✓	0/5	all ≈ 0.046
attention (rotary)	4	✓	6/10	mean 0.84; range 0.28–1.00

more parameters: the wall is a training-coverage gap, and training in this regime is a lottery whose rate the curriculum's shape controls. One boundary is added there and not visible here: every converted model that solves its trained token band reads chance on held-out token bands, so what transfers is a binding circuit over the symbols it trained on.

Lever attribution. At five seeds the picture is a family of lock-in lotteries whose rates the training signal shifts. Dense supervision alone can solve the task (one seed of five reaches 1.000) but has the lowest lock-in rate (1/5). The curriculum raises the rate (3/5 alone; 4/5 with dense supervision), consistent with the mechanism story (training across gaps builds the selective-write policy outward from the solvable adjacent case), and the causal cell locks in at a similar rate under the same recipe (3/5)—directional parity, consistent with the collision replication (§7). No pairwise rate contrast reaches significance at n=5, so we state the attribution as a directionally consistent rate shift rather than a settled lever.

Two boundaries. (i) The attention ceiling under the curriculum requires position generalization—and with it, exists. The absolute-position toy baseline fails to generalize across table positions when the layout shifts (robust across learning rates and seeds, 7/7 collapse at ≈0.046). Replacing its learned absolute table with rotary positions—whose attention scores are provably shift-invariant—restores the ceiling: rotary attention solves the fixed-layout wall (1.000) and learns under the shifting curriculum (6/10 lock-in lexical, 7/10 collision-ramp, with most misses graceful rather than collapsed). The absolute-position collapse is thereby confirmed as a position-generalization artifact, not an attention limitation. The shaped-collision recurrent cells' lock-in rate (9/10) is statistically indistinguishable from rotary attention's (7/10) at these n; the regimes are comparable, not ordered. (ii) The fix has a length boundary, which shaping re-opens—stochastically. Width scales cleanly: at d=128 (16× state) the L=64 collision-plus-curriculum recipe reaches 1.000 unchanged. Length under the uniform-gap curriculum does not: at L=256 (120 colliding decoy pairs vs. 24) every uniform cell is at chance—at both widths, at 4× the step budget, at two learning rates. A shaped (progressive-ramp) curriculum re-opens it: at five seeds the ramp locks in 4/5 (per-seed 0.052/0.994/0.996/0.868/1.000) against 0/9 pooled uniform trials across widths, budgets, and learning rates—$p\approx0.005$ (Fisher, one-sided; the uniform trials pool several configurations). This is the one curriculum-shape contrast in this paper that clears significance. So L=256 is not an architectural interference ceiling; it is the lock-in-lottery phenomenon at a longer horizon, with curriculum shape as a reliable rate lever, and the residual lock-out motivating success-gated ramps. A further 4× in length then exhausts the open-loop ramp outright: at L=512 every ramp trial sits at chance (0/6; bidirectional 0.020/0.020/0.019, causal 0.023/0.018/0.019) against 4/5 at L=256. Length is thus itself a rate-killer for a time-based schedule, which advances on a clock whether or not the model is keeping up—the failure a closed-loop schedule removes. A companion paper shows exactly that fix working at conversion scale (Boesch & Wee, 2026); the gated schedule at L=512 is its direct toy test. Mean-level results should be quoted as established at L=64 and, in length, as lock-in rates only.

7. Bidirectional Denoisers Read Twice for Free

Just-Read-Twice (Arora et al., 2024b) showed that recurrent LMs recover most of the recall gap when the query precedes the context—either by repeating the prompt (JRT-Prompt) or with a bespoke prefix-LM architecture (JRT-RNN). We observe that bidirectional masked-denoiser cells have this property natively: the backward stream reads the query before the haystack, by construction, in every bidirectional diffusion-LM denoiser (Sahoo et al., 2024; Singh et al., 2025). No prompt repetition, no bespoke architecture. This section isolates whether that built-in second read is a real mechanism, what circuit implements it, and what it requires.

The discriminator. In the lexical haystack (§6) the causal cell also lifts under the curriculum (3/5 seeds lock in, mean 0.72; Table 5), because distractors are lexically distinguishable from table keys—a causal lexical write gate is learnable in principle, so directionality is a margin there, not a mechanism. The collision-key variant kills that route: decoys reuse the table's own keys, so no lexical gate can separate a distractor write from a table write. Theory going in: an input-conditioned causal write gate must overwrite on key reuse (retaining garbage → chance); the backward stream reads the query first, and its reverse-scan overwrite retains the correct first-in-forward binding.

Reads (Table 6).

1. No directional margin at ten seeds. Both arms are lock-in lotteries (bidirectional locks in 3/10, causal 1/10—and the causal count is threshold-sensitive: at a 0.6 criterion both arms lock in 3/10). Means differ by +0.082 (p=0.63). A two-seed read of the same arms (1.000/0.933 bidirectional vs. 0.73 causal) had suggested a large margin; those were the two luckiest draws. "Bidirectional denoisers get Just-Read-Twice for free" is not established as a differential claim at this power.

2. What does survive: solvability, in both directions. Some seeds of both arms solve collision-key retrieval outright (bidirectional max 1.000; causal max 0.844). The causal existence result is itself informative—it robustly contradicts the single-cell theory (an input-conditioned write gate "must" overwrite on key reuse), strengthening the depth-as-novelty-gate hypothesis: layer-1 state can encode "this key is already bound," making layer-2's gate effectively state-conditioned. We report that as a hypothesis consistent with, but not isolated by, these data.

3. The depth requirement is robust. At one layer both arms collapse to chance. The mechanistic reason is exact: the masked answer occupies the last position—the first token of the backward scan—so a single backward pass has absorbed nothing by the time it reaches the answer slot; query-first information exists only at earlier positions and must be routed to the answer by a second layer's forward stream (mark-and-route). Whatever circuit solves this task, it needs two layers.

4. Lock-in is the phenomenon to explain. The same bimodality appears in the shaped-curriculum length extension (§6: ramp seeds solve L=256 at 0.994 or lock out at 0.05). Curriculum training on interference tasks converges stochastically; the right statistic is a lock-in rate, mean accuracy is misleading, and differential claims between arms require far more than ten seeds at these rates (3/10 vs. 1/10 is $p\approx$0.58 by Fisher exact). The convolution-free arm has never locked in (0/10 under the uniform recipe, 0/10 under the shaped ramp), consistent with the reversed-order-binding account of the convolution's role, but the same power caveat applies.

Implication for diffusion LMs. The architectural observation stands: bidirectional denoisers possess query-first reading natively (the JRT property causal recurrent LMs must engineer via prompt repetition or prefix-LM encoders), and any circuit exploiting it is necessarily $\geq$2 layers (mark-and-route). What these data do not show is that the native property confers a measurable advantage over a causal cell trained the same way: at ten seeds the directional difference is indistinguishable from the lock-in lottery. The natural follow-up—does shaped training that stabilizes lock-in also separate the directions?—is negative: under the progressive-ramp curriculum, collision lock-in rises from 3/10 to 9/10 (bidirectional) and from 1/10 to 9/10 (causal; $p\approx 6\times 10^{-4}$ vs. its uniform reference), at exact directional parity (p=0.76). The question is closed, negatively: in this task family, query-first reading confers no measurable recall advantage even when training is stabilized. The durable finding is that curriculum shape is a powerful, direction-agnostic lock-in stabilizer—the second shape contrast in this paper to clear significance.

8. Guardrail: Arming Is Free on State Tracking

Does adding a local (finite-window) convolution cost the recurrence anything on the axis where recurrences are supposed to shine—state tracking? We compare armed_gdn vs. gated_deltanet on the S_5 running-product guardrail (§3.3).

We frame this strictly as learnability at fixed depth, not as a circuit-complexity statement: accuracy falls steeply with composition depth for both arms, and nothing here bears on asymptotic separations (Merrill et al., 2024; Grazzi et al., 2025). The design question was narrow—does the recall fix tax the state-tracking axis?—and the answer is no.

Table 6. Collision-key discriminator, hardened (L=64, dense-4 + curriculum, 2 layers; n=10 seeds, rebinding-controlled—the armed arms' rebound sits at the value-marginal floor, while the convolution-free arm's rebound equals its accuracy: it never binds at all). Per-seed accuracies are bimodal: most seeds lock out, some solve outright, so we report lock-in rates (seeds > 0.8) alongside means. The bidirectional–causal mean difference is +0.082 (p=0.63, permutation): no directional difference is established at this power. Floors: value-marginal ≈ 0.019; strongest non-binding positional strategy ≈ 0.26.

arm (L=64 collision)	lock-in	mean	per-seed range
armed_gdn (bidirectional)	3/10	0.383	0.023–1.000
armed_gdn_fwd (causal)	1/10	0.301	0.036–0.844
gated_deltanet (bidir, no conv)	0/10	0.128	0.017–0.238
1-layer: bidir 0.031/0.022; causal 0.042/0.046 — both chance			

Table 7. S_5 guardrail (d=128; chance ≈ 0.008; internally comparable only). L=8/L=16 are the ten-seed hardened rows (mean [min, max], permutation p); L=2/L=4 remain 2-seed context. At full power the armed cell is significantly better at both probed depths: arming for recall does not trade away state tracking. The result is cell-family-general: an armed RWKV-7 cell also beats its unarmed counterpart at both depths (+0.053, p=0.004 at L=8; +0.005, $p<10^{-4}$ at L=16) and is statistically indistinguishable from armed armed_gdn (p=0.60/0.10).

S_5 depth	armed_gdn	gated_deltanet	p
L=2 (n=2)	1.000	0.975	—
L=4 (n=2)	0.518	0.471	—
L=8 (n=10)	0.214 [0.143, 0.261]	0.152 [0.132, 0.240]	0.0044
L=16 (n=10)	0.087 [0.072, 0.128]	0.071 [0.070, 0.073]	$< 10^{-4}$

9. Limitations and Threats to Validity

1. The Mamba-2 reference under-trains the official kernel. mamba2_ref reaches 0.592 at 10 seeds (0.641 at 3) where the official fused Mamba-2 reached 0.884 at the same matched state. The within-harness factorial is internally valid (all cells, one harness, one budget), but absolute margins against a real Mamba are softer, and the with-conv transition gap is measured ablation-grade within one cell (+0.034, §4) so it does not depend on this comparator. Re-running armed_gdn on Ampere in bf16 through the same probe script and protocol gives 0.990 mean at K=32 against the official kernel's 0.884—a +0.11 margin—but the two sides were run in separate sessions and the official arm was not per-arm tuned. A paired, tuned measurement in one session remains open.

2. Toy scale, single task family. $d \in \{32, 128\}$, one MQAR configuration per setting, synthetic tasks. The decomposition's magnitudes (+0.5 conv; +0.3 transition without conv, +0.03 with it; decay unresolved) are specific to constrained state; at ample state everything saturates and the knobs stop mattering. Claims should be read as mechanism isolation, not deployment guidance.

3. Learning-rate sensitivity. Okpekpe & Orvieto (2025) show per-arm LR tuning can rescue no-conv Mamba recall; our sweep (§4.2) confirms this for the delta family (tuned no-conv gated_deltanet 0.870; re-tuning the armed arms too narrows the delta-family gap to $\approx$+0.13) and not for our diagonal cell (0.172 at its best lr). The sweep spans one decade at 5 seeds, and per-arm tuning of the official Mamba-2 kernel is not yet done (item 1).

4. The attention control is valid only where reported. Running attention (absolute and rotary) at a healthy head dimension (2 heads, d=32) through the factorial's matched one-layer config yields collapse at every K ($\approx$0.11–0.17) with the rebinding control equal to eval accuracy—the fingerprint of never reading bindings. This is structural: MQAR is an induction-head task and induction circuits require two attention layers (Olsson et al., 2022), while the recurrent cells solve it at depth one by writing bindings into state. A depth-matched attention row therefore cannot exist at L=1; the valid attention ceilings live in the two-layer regimes, where we report them (fixed wall 1.000; lexical curriculum 6/10; collision-ramp 7/10; L=256 ramp 4/5—parity with the armed cell's 4/5). This paper makes no attention-superiority claims beyond those settings.

5. Cross-family magnitude comparisons are indicative. mamba2_ref's decay is scalar-per-head; the RWKV-7 reference uses vector decay; gated_deltanet scalar decay. Within-family toggles are exact; cross-family magnitude comparisons

(e.g., conv effect size in delta vs. diagonal families) carry a parameterization caveat.

6. Interpretation boundaries we respect. No rank-1-vs-diagonal class claim (§4.3 kills it); no claim that decay helps state-tracking (underpowered); no length extrapolation of the curriculum fix beyond L=64; no circuit-complexity claims from the S_5 guardrail; no bidirectional-vs-causal advantage claim under collision (n.s. at 10 seeds); the depth-novelty-gating account of the causal arm's solvability is a hypothesis.

10. Discussion

What "recurrent models are bad at recall" decomposes into. At matched state: a missing convolution (large, fixable for free in state terms), a transition effect (real, within-cell), a decay tax (real, scalar-gate-specific), an interference failure under sparse supervision (fixable by curriculum, within a length boundary), and a directionality property (bidirectional denoisers get query-first reading natively). None of these is "the recurrence cannot bind."

When is a hybrid actually needed? The results rescope the standard "add attention layers for retrieval" move. Two of the classic triggers dissolve under cheap interventions: the unarmed-cell recall deficit (add the convolution) and the within-capacity haystack wall (train with a distance curriculum). What remains as genuine hybrid territory: loads beyond state capacity (the graceful K-limit), lengths beyond the curriculum boundary ($L \geq 256$ here, pending shaped curricula), and query-conditioned regimes a fixed-state writer cannot anticipate. A hybrid decision made after arming and curriculum is a different, and smaller, decision.

For recurrent diffusion LMs. The JRT-for-free result gives bidirectional recurrent denoisers (Singh et al., 2025; Lin et al., 2026) a principled recall story that causal recurrent LMs lack: the objective itself buys the second read. The requirements are concrete—two layers minimum and a short convolution in the backward stream—and they are design guidance for this model class, not folklore. Whether the mechanism carries from toy probes to trained diffusion LMs at scale is the natural next experiment.

A conjecture, and a design question. Within this study, one apparent architectural wall dissolved entirely into a training-distribution problem, and convergent evidence points the same way for recurrent recall and length generalization (Okpekpe & Orvieto, 2025; Buitrago Ruiz & Gu, 2025; Blouir et al., 2024); we state the strong form as a conjecture worth falsifying—within the capacity regime of a fixed-state recurrence, apparent retrieval walls are training-coverage gaps—with the L=256 boundary as its live test against the alternative (a genuine interference ceiling of the two-layer circuit). Separately, decay costs -0.32 on recall here while showing no measured state-tracking benefit at our power (p=0.65); only if a benefit emerges under better-powered tests does context-modulated forgetting (Kimi Team, 2025) become the interesting design axis.

Reproducibility Statement

All cells, tasks, and controls run in one open harness (pure-PyTorch reference cells; a state-matched, kernel-free Mamba-2 comparator; fp32 paths) on commodity GPUs, including pre-Volta hardware via an atomic-free two-pass Triton backward (Appendix A). Appendix B lists the exact commands for every table. Code and result JSONs are publicly available at https://github.com/JIBSIL/dualgoose.

Impact Statement

This paper presents controlled, small-scale measurements aimed at advancing the scientific understanding of sequence-model architectures: which ingredients of efficient recurrent cells support in-context recall, and when training rather than architecture is the binding constraint. The main downstream impact, if the findings transfer to scale, is more capable linear-time language models at lower inference cost and energy. We see no societal consequences specific to this work beyond those generic to improving machine learning methods.

References

Arora, A., Rathi, N., Selvam, N. R., Csordás, R., Jurafsky, D., and Potts, C. Mechanistic evaluation of transformers and state space models. arXiv preprint arXiv:2505.15105, 2025.

Arora, S., Eyuboglu, S., Timalsina, A., Johnson, I., Poli, M., Zou, J., Rudra, A., and Ré, C. Zoology: Measuring and improving recall in efficient language models. arXiv preprint arXiv:2312.04927, 2023.

Arora, S., Eyuboglu, S., Zhang, M., Timalsina, A., Alberti, S., Zinsley, D., Zou, J., Rudra, A., and Ré, C. Simple linear attention language models balance the recall-throughput tradeoff. arXiv preprint arXiv:2402.18668, 2024a.

Arora, S., Timalsina, A., Singhal, A., Eyuboglu, S., Zhao, X., Rao, A., Rudra, A., and Ré, C. Just read twice: Closing the recall gap for recurrent language models. arXiv preprint arXiv:2407.05483, 2024b.

Austin, J., Johnson, D. D., Ho, J., Tarlow, D., and van den Berg, R. Structured denoising diffusion models in discrete state-spaces. arXiv preprint arXiv:2107.03006, 2021.

Bachmann, G. and Nagarajan, V. The pitfalls of next-token prediction. arXiv preprint arXiv:2403.06963, 2024.

Ben-Kish, A., Zimerman, I., Abu-Hussein, S., Cohen, N., Globerson, A., Wolf, L., and Giryes, R. DeciMamba: Exploring the length extrapolation potential of Mamba. arXiv preprint arXiv:2406.14528, 2024.

Bengio, Y., Louradour, J., Collobert, R., and Weston, J. Curriculum learning. Proceedings of ICML, 2009.

Blouir, S., Smith, J. T., and Anastasopoulos, A. Birdie: Advancing state space models with reward-driven objectives and curricula. In Proceedings of EMNLP, 2024.

Boesch, J. and Wee, A. DreamingGoose: Staged distillation from autoregressive transformers to bidirectional recurrent diffusion language models. Companion preprint, 2026.

Buitrago Ruiz, R. and Gu, A. Understanding and improving length generalization in recurrent models. arXiv preprint arXiv:2507.02782, 2025.

Chen, Y., Zhang, X., Hu, S., Han, X., Liu, Z., and Sun, M. Stuffed Mamba: State collapse and state capacity of RNN-based long-context modeling. arXiv preprint arXiv:2410.07145, 2024.

Dao, T. and Gu, A. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. arXiv preprint arXiv:2405.21060, 2024.

Delétang, G., Ruoss, A., Grau-Moya, J., Genewein, T., Wenliang, L. K., Catt, E., Cundy, C., Hutter, M., Legg, S., Veness, J., and Ortega, P. A. Neural networks and the Chomsky hierarchy. arXiv preprint arXiv:2207.02098, 2023.

Grazzi, R., Siems, J., Franke, J. K., Zela, A., Hutter, F., and Pontil, M. Unlocking state-tracking in linear RNNs through negative eigenvalues. arXiv preprint arXiv:2411.12537, 2025.

Gu, A. and Dao, T. Mamba: Linear-time sequence modeling with selective state spaces. arXiv preprint arXiv:2312.00752, 2023.

Hewitt, J. and Liang, P. Designing and interpreting probes with control tasks. arXiv preprint arXiv:1909.03368, 2019.

Holtzman, A., West, P., Shwartz, V., Choi, Y., and Zettlemoyer, L. Surface form competition: Why the highest probability answer isn't always right. arXiv preprint arXiv:2104.08315, 2021.

Hsieh, C.-P., Sun, S., Kriman, S., Acharya, S., Rekesh, D., Jia, F., Zhang, Y., and Ginsburg, B. RULER: What's the real context size of your long-context language models? arXiv preprint arXiv:2404.06654, 2024.

Jelassi, S., Brandfonbrener, D., Kakade, S. M., and Malach, E. Repeat after me: Transformers are better than state space models at copying. arXiv preprint arXiv:2402.01032, 2024.

Katharopoulos, A., Vyas, A., Pappas, N., and Fleuret, F. Transformers are RNNs: Fast autoregressive transformers with linear attention. arXiv preprint arXiv:2006.16236, 2020.

Kimi Team. Kimi linear: An expressive, efficient attention architecture. arXiv preprint arXiv:2510.26692, 2025.

Lin, K., Luo, Y., Su, Z., Song, Y., and Rao, A. Triplet-block diffusion RWKV. arXiv preprint arXiv:2605.25969, 2026.

Liu, B., Ash, J. T., Goel, S., Krishnamurthy, A., and Zhang, C. Transformers learn shortcuts to automata. arXiv preprint arXiv:2210.10749, 2023.

Merrill, W., Petty, J., and Sabharwal, A. The illusion of state in state-space models. arXiv preprint arXiv:2404.08819, 2024.

Okpekpe, D. and Orvieto, A. Revisiting associative recall in modern recurrent models. arXiv preprint arXiv:2508.19029, 2025.

Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., Mann, B., Askell, A., Bai, Y., Chen, A., et al. In-context learning and induction heads. Transformer Circuits Thread, 2022.

Parnichkun, R. N., Tumma, N., Thomas, A. W., Moro, A., An, Q., Suzuki, T., Yamashita, A., Poli, M., and Massaroli, S. Quantifying memory utilization with effective state-size. arXiv preprint arXiv:2504.19561, 2025.

Peng, B., Zhang, R., Goldstein, D., et al. RWKV-7 "Goose" with expressive dynamic state evolution. arXiv preprint arXiv:2503.14456, 2025.

Sahoo, S. S., Arriola, M., Schiff, Y., Gokaslan, A., Marroquin, E., Chiu, J. T., Rush, A., and Kuleshov, V. Simple and effective masked diffusion language models. arXiv preprint arXiv:2406.07524, 2024.

Schlag, I., Irie, K., and Schmidhuber, J. Linear transformers are secretly fast weight programmers. arXiv preprint arXiv:2102.11174, 2021.

Singh, V., Ostapenko, O., Noël, P.-A., Belilovsky, E., and Scholak, T. DiffuMamba: High-throughput diffusion LMs with Mamba backbone. arXiv preprint arXiv:2511.15927, 2025.

Sridhar, A. and Johansen, A. Echo: KV-cache-free associative recall with spectral Koopman operators. arXiv preprint arXiv:2605.06997, 2026.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. Advances in Neural Information Processing Systems, 2017.

Yang, S., Kautz, J., and Hatamizadeh, A. Gated delta networks: Improving Mamba2 with delta rule. arXiv preprint arXiv:2412.06464, 2024a.

Yang, S., Wang, B., Zhang, Y., Shen, Y., and Kim, Y. Parallelizing linear transformers with the delta rule over sequence length. arXiv preprint arXiv:2406.06484, 2024b.

A. The Commodity-Hardware Harness

Three engineering choices make the factorial cheap to replicate:

Kernel-free cells. Every arm has a pure-PyTorch reference implementation; the Mamba-2 comparator (TinyMamba2RefSSM) implements the SSD recurrence with an explicit d_{state} and requires neither Triton nor mamba_ssm, so the full factorial executes on any CUDA GPU (and, slowly, CPU). The kernel_size=1 setting neutralizes its convolution for the toggle.

Pre-Volta support. The default Triton backward for the RWKV-7-style scan emits .acq_rel atomics that ptxas rejects below sm_70. An atomic-free two-pass backward reproduces the reference scan's forward and gradients and runs on Pascal (sm_61) at ≈26× the pure-PyTorch reference throughput (≈2,980 vs. ≈113 tok/s at d=256/6 layers/ctx 512), with an –fp32 flag for hardware without bf16. The headline experiments in this paper were run on a single GTX 1070.

Determinism and floors. Runs are seed-pinned and resumable; every task ships its floor calculators (value-marginal, no-binding, table floor $1/K$, positional). The rebinding control is validated by a self-test (–selftest) that confirms the derangement logic.

B. Reproduction Commands

Constrained-state factorial and K-sweep (Tables 1, 2). –fp32 is required on pre-bf16 (Pascal) hardware and may be dropped on Ampere and newer; it applies to every command below except dg_c9_harden.py, which has no such flag:

```
python scripts/mqar_probe.py –models armed_gdn gated_deltanet deltanet
    dg_rank_ref dg_diag_ref mamba2_ref mamba2_ref_noconv attention
    –d-model 32 –layers 1 –mamba2-d-state 16 –ks 8 16 32 –seeds 0 1 2 –fp32
```

Hardened transition ablation with rebinding control (Table 3):

```
python scripts/dg_c9_harden.py –ks 16 32 –seeds 0 1 2 3 4 5 6 7 8 9
python scripts/dg_c9_harden.py –selftest
```

Haystack wall (Table 4):

```
python scripts/long_retrieval_probe.py –models armed_gdn dg_rank_ref mamba2_ref attention
    –seq-lens 64 128 256 512 –pairs 4 –seeds 0 1 –heads 2 –mamba2-d-state 16 –fp32
```

Curriculum and collision-key discriminator (Tables 5, 6); add –collision for the discriminator:

```
python scripts/long_retrieval_probe.py –models armed_gdn armed_gdn_fwd attention
    –seq-lens 64 –pairs 4 –seeds 0 –steps 3000 –batch 256 –layers 2 –heads 2
    –dense 4 –curriculum –fp32
```

S_5 guardrail (Table 7):

```
python scripts/state_tracking_probe.py –models armed_gdn gated_deltanet
    –group-n 5 –ls 8 16 –d-model 128 –seeds 0 1 2 3 4 5 6 7 8 9 –fp32
```

C. Floors and Controls, Stated

For each task we report: (i) the value-marginal (probability of the correct value under the answer distribution: $1/26 \approx 0.038$ for MQAR; $1/54 \approx 0.019$ for haystack/collision; $1/120 \approx 0.008$ for S_5, exact because generators are uniform over the full group); (ii) the strongest no-binding strategy (MQAR: emit a random table value, ≈0.068;

collision: emit a random table-position token, $\approx$0.26); (iii) the table floor $1/K$ for the rebinding control. No headline number in this paper is within $10\times$ of its floor except where explicitly marked as “chance.” The S_5 probe additionally masks all product slots simultaneously and uses single-token answers, making prefix-completion leakage structurally impossible.